

Protocol for whole shotgun metagenomics pipeline for the study of mice digestive microbiota

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DNA extraction, high throughput sequencing, read mapping and bioinformatical analysis to determine Metagenomic Species Pangenome (MSP) were performed for the study of the mice microbiota.

DNA extraction and high throughput sequencing. Frozen faecal materials were aliquoted to ≤ 250 mg and DNA extraction was performed following the procedure previously described in [dx.doi.org/10.17504/protocols.io.dm6gpjm11gzp/v1](https://doi.org/10.17504/protocols.io.dm6gpjm11gzp/v1), with the following modifications. 250 μ L guanidium thiocyanate, 40 μ L N-lauroyl sarcosine (10 % solution) and 500 μ L N-lauroyl sarcosine (5 % solution in PBS 1X) were added to each frozen mouse faecal sample, which were then homogenised with a toothpick, vortexed and transferred to a deep-well plate containing 400 μ L of 0.1 mm glass beads (not in suspension). Subsequently, the sample plate was incubated at 70°C in a thermomixer for one hour, with stirring at 1,400 rpm. Following centrifugation of the plate at 3,486 $\times$ g for a period of five minutes, the lysate was collected in a new plate. The pellet was then washed with 500 μ L of TENP (50 mM Tris-HCL 20 mM EDTA 10 mM NaCl, saturated with PVPP). The plate was vortexed and centrifuged at 3,486 $\times$ g for five minutes, after which the recovered lysate was pooled with the previous one. Finally, the final lysate was centrifuged for 10 minutes at 3,486 $\times$ g, after which 800 μ L were collected in a new plate. This plate was employed for purification with magnetic beads on the QIASymphony. The utilised protocol has been designed for MGP with the QIAGEN DSP Virus/Pathogen kit. DNA was quantified using Qubit Fluorometric Quantitation (ThermoFisher Scientific, Waltham, US) and qualified using DNA size profiling on a Fragment Analyzer (Agilent Technologies, Santa Clara, US). One μ g of high molecular weight DNA (>10 kbp) was used to build the library. Shearing of DNA into fragments of approximately 150 bp was performed using an ultrasonicator (Covaris, Woburn, US) and DNA fragment library construction was performed using the Ion Plus Fragment Library and Ion Xpress Barcode Adapters Kits (ThermoFisher Scientific, Waltham, US). Purified and amplified DNA fragment libraries were sequenced using the Ion Proton Sequencer (ThermoFisher Scientific, Waltham, US), with a minimum of 20 million high-quality 150 bp reads generated per library [1].

Read Mapping. Quality filtered to remove any low-quality sequences using Alentrimmer software [2] and potential host-related reads using Bowtie2 [3]. Resulting high-quality reads were mapped onto the 5 million gene integrated reference catalog of the Murine Intestinal Microbiota Integrated Catalog v2 (MIMIC2) [4] using the METEOR software suite [5]. Read mapping was performed in a two-step procedure, using an identity threshold of 95% to the reference gene catalog with Bowtie2 [3]. First, unique mapped reads were attributed to their corresponding genes. Second, shared reads were weighted according to the ratio of unique mapping counts. A downsizing procedure was performed to normalize the gene counts between samples by randomly selecting a subset of reads depending on the sequencing depth (usually ≥ 10 M reads for an average 20M reads depth sequencing). The gene abundance table was then normalized using the FPKM strategy and analysed using MetaOMineR (*momr*) R package (<https://cran.r-project.org/web/packages/momr/index.html>) [6].

MSP microbial species determination. MSPs were used to quantify species associated to the 5.0 million gene MIMIC2 reference catalogue [7, 8]. MSPs are clusters of co-abundant genes (min size ≥ 500 genes) that likely belong to the same microbial species, reconstructed from the 5 million genes catalogue into 1252 MSPs. MSP abundance profiles were calculated as the mean abundance of 100 marker genes, defined as the robust centroids of each MSP cluster. A threshold of 10% of the marker genes was applied as MSP detection limit. Taxonomical annotation was performed using GDTB R07-RS207 [9].

Assessment of the microbial functional potentials. To determine functional potential of the gut microbiota at the module level, we used an INRAE pipeline described in Thirion et al. [10]. Three databases were used to predict gene functions: Kyoto Encyclopedia of Genes and Genomes (KEGG) [11], eggNOG database (version 3.0) [12] and TIGRFAM (version 15.0) [13]. First, genes of the 5 million genes catalogue were annotated using KEGG107 database using Diamond [14] and further clustered into functional pathway modules according to KEGG (Kyoto Encyclopaedia of Genes and Genomes) Orthology (KO) groups, Gut Metabolic Modules (GMM), and Gut Brain Modules (GBM). Second, KEGG, GMM [15] and GBM modules [16] were reconstructed in each MSP using their reaction pathways based on their detected annotated KO genes. GMM and GBM functional modules were selected because they are specific to gut bacterial and gut-brain axis functions. For each pair of MSP/mouse, the completeness of any given functional modules was calculated by considering the set of genes detected in the MSP of each mouse and the MSP completeness in each mouse. For a given MSP in a specific mouse, completeness of the modules was corrected by the abundance of the MSP. After correction, functional modules in each MSP/mouse were considered as complete if at least 90% of the involved reactions were detected. Abundance of functional modules in each sample was computed as the sum of the MSP abundances containing the complete functional module.

References

1. Meslier V, Quinquis B, Da Silva K, Plaza Oñate F, Pons N, Roume H, Podar M, Almeida M. Benchmarking second and third-generation sequencing platforms for microbial metagenomics. *Sci Data*. 2022 Nov 11;9(1):694. doi: 10.1038/s41597-022-01762-z. PMID: 36369227; PMCID: PMC9652401.
2. Criscuolo, A.; Brisse, S. AlienTrimmer: A tool to quickly and accurately trim off multiple short contaminant sequences from high-throughput sequencing reads. *Genomics* 2013, 102, 500–506
3. Langmead, B.; Salzberg, S.L. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* 2012, 9, 357–359.
4. Plaza-Onate, Florian; Gitton-Quent, Oscar; Almeida, Mathieu; Le Chatelier, Emmanuelle, 2021, "MIMIC2: Murine Intestinal Microbiota Integrated Catalog v2", <https://doi.org/10.15454/L11MXM>, Recherche Data Gouv, V5, UNF:6:K30sRn45IbOJc6WU7dPAc== [fileUNF]
5. Pons N, Batto J-M, Kennedy S, Almeida M, Boumezeur F, Moumen B, et al. (2010): METEOR -a platform for quantitative metagenomic profiling of complex ecosystems. Retrieved from <https://forgemia.inra.fr/metagenopolis/meteor>
6. Le Chatelier E, Prifti E. Mining Metaomics Data In R. Retrived from <https://forgemia.inra.fr/metagenopolis/momr>
7. Nielsen, H.B.; Almeida, M.; Juncker, A.S.; Rasmussen, S.; Li, J.; Sunagawa, S.; Plichta, D.R.; Gautier, L.; Pedersen, A.G.; Le Chatelier, E.; et al. Identification and assembly of genomes and genetic elements in complex metagenomic samples without using reference genomes. *Nat. Biotechnol.* 2014, 32, 822–828

8. Plaza-Oñate, F.; Le Chatelier, E.; Almeida, M.; Cervino, A.C.L.; Gauthier, F.; Magoulès, F.; Ehrlich, S.D.; Pichaud, M. MSPminer: Abundance-based reconstitution of microbial pan-genomes from shotgun metagenomic data. *Bioinformatics* 2019, 35, 1544–1552
9. Parks, D.H., et al. (2021). "GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy." *Nucleic Acids Research*, <https://doi.org/10.1093/nar/gkab776>
10. Thirion, F., Speyer, H., Hansen, T. H., Nielsen, T., Fan, Y., Le Chatelier, E., Fromentin, S., Berland, M., Plaza Oñate, F., Pons, N., Galleron, N., Levenez, F., Markó, L., Birkner, T., Jørgensen, T., Forslund, S. K., Vestergaard, H., Hansen, T., Nordentoft, M., ... Ehrlich, S. D. (2023). Alteration of Gut Microbiome in Patients With Schizophrenia Indicates Links Between Bacterial Tyrosine Biosynthesis and Cognitive Dysfunction. *Biological Psychiatry Global Open Science*, 3(2), 283-291. <https://doi.org/10.1016/j.bpsgos.2022.01.009>
11. Kanehisa M, Goto S. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Res.* 2000;28(1):27–30. doi:10.1093/nar/28.1.27
12. Huerta-Cepas J, Szklarczyk D, Forslund K, et al. EGGNOG 4.5: a hierarchical orthology framework with improved functional annotations for eukaryotic, prokaryotic and viral sequences. *Nucleic Acids Res.* 2016;44(D1):D286–93. doi:10.1093/nar/gkv1248
13. Haft DH. TIGRFAMs: a protein family resource for the functional identification of proteins. *Nucleic Acids Res.* 2001;29(1):41–43. doi:10.1093/nar/29.1.
14. Buchfink B, Xie C, Huson DH. Fast and sensitive protein alignment using DIAMOND. *Nat Methods.* 2015;12(1):59–60. doi:10.1038/nmeth.3176
15. Vieira-Silva S, Falony G, Darzi Y, et al. Species-function relationships shape ecological properties of the human gut microbiome. *Nat Microbiol.* 2016;1(8):16088. doi:10.1038/nmicrobiol.2016.88
16. Valles-Colomer M, Falony G, Darzi Y, et al. The neuroactive potential of the human gut microbiota in quality of life and depression. *Nat Microbiol.* 2019;Apr(4):623–632. doi:10.1038/s41564-018-0337-x